2.2.1 The heart and monitoring heart function

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	the need for a mass transport system in mammals	To include references to a high basal metabolic rate, being multicellular and the significance of surface area to volume ratio. *M0.3, M4.1*
(b)	**(i)** the internal and external structure of the mammalian heart **(ii)** the examination, dissection and drawing of the mammalian heart	**PAG2** HSW3, HSW4, HSW5, HSW6, HSW8
(c)	the cardiac cycle	To include the role of the valves and the pressure changes occurring in the heart and associated vessels.
(d)	how heart action is initiated and co-ordinated	To include the roles of the sino-atrial node (SAN), atrio-ventricular node (AVN), purkyne tissue and the myogenic nature of cardiac muscle (no detail on hormonal and nervous control is required at AS level but at A level this is covered in 5.3.1(b)).
(e)	practical investigation(s) into the factors affecting heart rate	To include the effect of exercise. *M0.1, M0.4, M1.2, M1.3, M1.11, M2.3, M2.4, M3.1, M3.2* **PAG10** HSW3, HSW4, HSW5, HSW6

(f)	the effect of heart rate on cardiac output	To include calculations based on heart rate and stroke volume. *M0.1, M0.4, M1.2, M1.3, M1.11, M2.3, M2.4, M3.1, M3.2* **PAG10** HSW3, HSW4, HSW5, HSW6
(g)	the measurement and interpretation of pulse rate, to include the generation of primary data and the use of secondary data	*M0.1, M0.4, M1.2, M1.3, M1.10, M1.11, M3.1, M3.2* **PAG10** HSW3, HSW4, HSW5, HSW6
(h)	the use and interpretation of an electrocardiogram (ECG)	To include tachycardia, bradycardia, S-T elevation and fibrillation. *M3.1* HSW3, HSW5
(i)	the emergency treatment given to a person suffering a suspected heart attack or cardiac arrest.	To include the first-aid treatment for a heart attack and cardiac arrest **AND** the use of a defibrillator following cardiac arrest.